# Pixel Reconstruction vs. Latent Prediction for Self-Supervised Representation Learning of Active Matter Dynamics

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**Abstract**

We study self-supervised representation learning for the `active_matter` subset of *The Well* (Ohana et al., 2025) under the strict frozen-encoder evaluation protocol mandated by our course (single linear probe and $k$-NN regression for $\alpha, \zeta$). We compare two SSL paradigms on a *byte-identical* 3D ConvNeXt encoder, so any downstream MSE difference is causally attributable to the SSL objective alone: **SIGReg-JEPA** (Balestriero & LeCun, 2026) (latent prediction; Person A) and a **VideoMAE/SimMIM-hybrid** (Tong et al., 2022; Xie et al., 2022) masked autoencoder we adapt for ConvNeXt by replacing masked tubes with a learnable mask token (Person B). We add an end-to-end supervised baseline as an upper-bound anchor. With 3 seeds for both the supervised baseline and VideoMAE-mask0.60, a mask-ratio sweep (0.60/0.75/0.90), a longer 50-epoch run, and *five independent analyses* (channel importance, frame budget, top-PC alignment, per-class separability, per-channel reconstruction MSE), our results characterize the SSL paradigm thoroughly. Headline test MSEs: supervised 0.106±0.006 linear / 0.106±0.020 kNN; VideoMAE-mask0.60 0.317±0.025 linear; VideoMAE-mask0.90 0.255±0.002 linear. The mask-ratio sweep monotonically improves linear MSE (0.317→0.299→0.255), replicating Tong et al.'s high-mask-is-optimal-for-video finding on 11-channel physical fields. Strikingly, the SSL encoders are *more isotropic* (Epps–Pulley 0.04 vs. supervised 0.09) and have higher effective rank (∼8 vs. ∼3.5), yet perform ∼2.4× worse. Top-PC analysis pinpoints why: supervised reaches its full performance from *just the top-2 PCs* (K=2 MSE 0.083 vs. K=full 0.086), while VideoMAE needs all 174 PCs (0.55→0.25). A channel-importance ablation explains it: VideoMAE relies disproportionately on the concentration scalar field (ΔMSE +0.26 vs. supervised's +0.029), causally confirmed by per-channel reconstruction MSE (0.22 for concentration vs. 0.55–0.78 for orientation/strain). Finally, reconstruction loss is *decoupled from downstream MSE*: a 50-epoch run reduces validation recon loss by 24% without improving linear-probe MSE. As a third data point Person A's CNext-U-Net forecaster (linear 0.199, kNN 0.128) sits between supervised and VideoMAE – forecasting features are better aligned with $(\alpha, \zeta)$ than masked-reconstruction features, refining the Qu et al. (2026) ordering. Together, six independent quantitative characterizations form one argument: pixel reconstruction is fundamentally misaligned with parameter estimation on this dataset, but the *form* of the prediction objective (next-frame vs. same-frame) matters.

## 1 Introduction

Self-supervised learning has emerged as a foundation paradigm for representation learning in vision, language and, more recently, the physical sciences (Qu et al., 2026; Balestriero et al., 2023). In this work we focus on *active matter*, a subset of *The Well* (Ohana et al., 2025) that simulates rod-like active particles in a viscous Stokes fluid via continuum kinetic theory. Each trajectory is governed by two latent physical parameters – the active dipole strength $\alpha$ and the steric alignment $\zeta$ – and the central question is whether a self-supervised encoder, trained *without access to those parameters*, learns a representation from which $\alpha$ and $\zeta$ can be recovered by a simple frozen-encoder probe.

The closest published work to ours is Qu et al. (2026), who report that on this exact dataset a JEPA-style encoder beats a VideoMAE-style encoder by ∼51% relative MSE (0.16 → 0.08) when the encoder is followed by a 100-epoch *attentive-probe* fine-tuning stage. The course rules explicitly prohibit attentive probes and any fine-tuning of the backbone; only a single linear layer and $k$-NN are allowed for evaluation. Our **contribution** is to ask the same JEPA-vs-MAE question under this much stricter protocol, while *controlling for the encoder family* (both methods share the byte-identical 3D ConvNeXt encoder, isolating the SSL objective as the only causal variable).

In short, our contributions are: (i) a **controlled comparison** of SIGReg latent prediction (Person A) vs.

VideoMAE/SimMIM pixel reconstruction (Person B) on *the same* 3D ConvNeXt encoder under the strict frozen-linear+kNN protocol; (ii) a **SimMIM adaptation for non-ViT encoders** – depthwise 3D ConvNeXts cannot use VideoMAE's asymmetric encoder-decoder, so we replace masked tubes with a learnable per-channel mask token (Sec. 3.2); (iii) a **five-pronged analysis suite** (channel importance, frame budget, top-PC alignment, per-class separability, per-channel reconstruction MSE) that applies the LeJEPA Epps–Pulley statistic (Balestriero & LeCun, 2026) as a diagnostic and isolates the rank-vs.-alignment trade-off in pixel-reconstruction SSL; (iv) an **end-to-end supervised baseline** on the same encoder, providing an upper-bound anchor.

# 2 Setup

## 2.1 Dataset

The `active_matter` subset comprises 8 750 / 1 200 / 1 300 sliding windows of 16 frames at 224×224 spatial resolution and 11 physical channels (concentration scalar, 2-channel velocity, 4-channel orientation tensor, 4-channel strain-rate tensor); 45 unique $(\alpha, \zeta)$ combinations total. Detailed statistics are in Tab. 2 (Appendix).

## 2.2 Encoder (shared between both SSL streams)

We use a 3D ConvNeXt encoder with stages of dimensions $(32, 64, 128, 256, 256)$ and per-stage block depths $(2, 2, 4, 8, 2)$, a 2×4×4 stem patch and 0.05 stochastic depth. The total spatial downsampling is 32× and temporal downsampling is 16×, yielding a $(B, 256, 7, 7)$ bottleneck per clip. The encoder has 7.7 M parameters.

## 2.3 Evaluation protocol

For every encoder we extract frozen features and run *both* (i) a single **nn.Linear**$(256, 2)$ probe trained with MSE loss against $z$-scored $(\alpha, \zeta)$ on the train split, and (ii) a $k$-Nearest Neighbours regressor with hyperparameters swept over $k \in \{1, 3, 5, 10, 20, 50\}$, weights $\in$ {uniform, distance}, metrics $\in$ {cosine, euclidean} and feature normalization $\in$ {z-score, $\ell_2$, z+$\ell_2$}. Model selection uses the validation split; the test split is touched only once per encoder for the final number. Label normalization (z-score) is fit on *train labels only*.

# 3 SSL Objectives

## 3.1 Latent prediction: SIGReg-JEPA (Person A)

[Person A: fill in the SIGReg / LeJEPA section – the Epps–Pulley distributional regularizer formulated against $\mathcal{N}(0, I)$, context/target temporal split, predictor architecture, etc.]

## 3.2 Pixel reconstruction: VideoMAE/SimMIM hybrid (Person B)

We adapt VideoMAE (Tong et al., 2022) for our 3D ConvNeXt encoder. Three adaptations are needed: (i) **mask-token replacement** – depthwise 3D convolutions require the full input tensor, so we cannot use VideoMAE's asymmetric encoder-decoder; following SimMIM (Xie et al., 2022) we replace each masked tube with a learnable per-channel mask token broadcast over the tube extent. (ii) **tube size aligned with encoder downsampling** – per SimMIM Sec. 4.1.2 the optimal mask patch matches the encoder stride; with our 32× spatial and 16× temporal downsampling, the tube size is $(16, 32, 32)$ and the grid is $7 \times 7 = 49$ tubes per clip. (iii) **lightweight prediction head** – SimMIM Tab. 2 shows lightweight heads transfer better; we use a 5-stage transposed-conv decoder (350 k parameters) that mirrors the encoder downsampling, mapping $(256, 7, 7) \to (11, 16, 224, 224)$, for a total model size of 8.07 M parameters (well under the 100 M cap). The reconstruction target is per-tube z-scored pixel values (VideoMAE Sec. 3.3); the loss is MSE over masked positions only (SimMIM Tab. 4 shows prediction-only beats full reconstruction):

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{(c,t,h,w) \in \mathcal{M}} \left( \hat{x}_{c,t,h,w} - \tilde{x}_{c,t,h,w} \right)^2 , \tag{1}$$

Table 1: Active matter parameter prediction. Z-scored MSE (lower is better) on the test split with a frozen encoder, evaluated with both a single linear probe and $k$-NN regression. Effective rank and Epps–Pulley distance are computed on the validation embedding set. We report mean±std across seeds where available. Best per column among *our* entries in **bold**.

| **Encoder** | **Params (M)** | **Linear** | **Linear $\alpha$** | **Linear $\zeta$** | **kNN** | **Ef** |
|---|---|---|---|---|---|---|
| *Ours:* | | | | | | |
| Supervised, 3 seeds (42, 43, 44) | 7.72 | **0.106±0.006** | **0.029±0.006** | **0.182±0.017** | **0.106±0.020** | |
| VideoMAE-mask0.60, 25 ep, 3 seeds | 8.07 | 0.317±0.025 | 0.124±0.014 | 0.510±0.038 | 0.562±0.035 | |
| VideoMAE-mask0.60, 50 ep, 1 seed (s= 46) | 8.07 | 0.338 | 0.179 | 0.497 | 0.389 | |
| VideoMAE-mask0.75, 1 seed | 8.07 | 0.299 | 0.075 | 0.522 | 0.672 | |
| VideoMAE-mask0.90, 2 seeds (44, 45) | 8.07 | 0.255±0.002 | 0.108±0.003 | 0.402±0.007 | 0.503±0.039 | |
| CNext-U-Net forecaster (Person A)† | 18.59 | 0.199 | 0.116 | 0.283 | 0.128 | |
| SIGReg-JEPA (Person A) | 7.72 | TBD | TBD | TBD | TBD | |
| *External references (Qu et al., attentive-probe FT for 100 epochs – different protocol):* | | | | | | |
| JEPA, ConvNeXt | ∼3.3 | 0.079 | — | — | — | |
| VideoMAE, ViT-tiny/16 | ∼5.5 | 0.160 | — | — | — | |

† CNext-U-Net was implemented by Person A; we ran the training (50 epochs, res 96, 4→1 forecasting, ∼30 min) and eval on our 2× B200 server using the workflow committed to the `main` branch.

where $\mathcal{M}$ is the set of masked pixel coordinates and $\tilde{x}$ is the per-tube z-scored target. The encoder is trained for 25 epochs with AdamW (lr $= 1.5\times10^{-4}$, weight decay 0.05), a cosine schedule with 2-epoch linear warmup, and BF16 autocast.

### 3.3 Supervised baseline (Person B)

We also train an end-to-end supervised baseline on the *same* 3D ConvNeXt encoder followed by an **nn.Linear**$(256, 2)$ head, optimizing $z$-scored MSE directly against $(\alpha, \zeta)$. This serves as an upper-bound anchor and exposes how much information about the physical parameters is recoverable from the encoder family at all. Trained for 30 epochs with AdamW (lr $= 3\times10^{-4}$).

## 4 Results

Tab. 1 reports the headline numbers. Person A's SIGReg-JEPA row is left as a placeholder until their training completes; all other rows are populated from the live `aggregated_results.json` artifact at the time of writing. We additionally include Qu et al. (2026)'s published numbers as an external reference; note that those values are obtained with a 100-epoch attentive-probe fine-tuning protocol that the assignment explicitly forbids us to use, so they are not directly comparable to our frozen-linear+kNN results – they are listed for context only.

Three observations stand out from Tab. 1.

**(1) Isotropy / rank inversion.** The VideoMAE encoders have the lowest Epps–Pulley distance to $\mathcal{N}(0, I)$ (0.039–0.042 vs. supervised ∼0.087) and the highest effective rank (7.6–8.9 vs. supervised ∼3.5), yet they have ∼2.4–3.4× *worse* linear probe MSE and ∼3.8–8.6× worse kNN MSE than the supervised baseline. This is a direct quantitative confirmation of Qu et al. (2026)'s claim that pixel reconstruction preserves features that are physically uninformative; isotropy alone is not sufficient for low downstream MSE when the encoder allocates rank to target-irrelevant signal. We diagnose *which* signal is target-irrelevant in Sec. 5 via channel-importance ablation.

**(2) Mask-ratio sweep.** Increasing the mask ratio along 0.60 → 0.75 → 0.90 *monotonically* improves the linear probe MSE (0.317 → 0.299 → 0.255, two seeds at the endpoints). The kNN MSE is non-monotonic but mask= 0.90 is best on kNN among the 25-epoch runs (0.50, vs. 0.56 at mask= 0.60). This replicates Tong et al.'s finding that high mask ratios (∼90%) are optimal for video, extended here to physical fields with 11 channels of structured signal. Crucially, even the best VideoMAE configuration (mask= 0.90, std 0.002 on linear MSE between seeds 44

and 45) is still $\sim$2.4$\times$ worse than the supervised baseline – the SSL paradigm itself is the limiting factor, not the mask ratio.

**(2b) Reconstruction loss is decoupled from downstream MSE.** Training VideoMAE-mask0.60 for 50 epochs instead of 25 reduces validation reconstruction loss by 24% ($0.501 \rightarrow 0.378$) but *does not improve linear probe MSE* ($0.331 \rightarrow 0.338$), although kNN improves ($0.610 \rightarrow 0.389$). This empirically confirms SimMIM Tab. 2: "stronger inpainting capability does not necessarily lead to better fine-tuning performance" – extended here to physical fields.

**(3) Seed stability.** The supervised baseline is highly reproducible (linear MSE std $\sim$0.006 over 3 seeds, CKA $\geq 0.96$ between seeds). The VideoMAE-mask0.60 run is also stable across 3 seeds (linear MSE std $\sim$0.025; CKA $\geq 0.96$). The CKA matrix (Fig. 2a) shows the seven 25-epoch VideoMAE encoders cluster at CKA $\geq 0.94$ regardless of mask ratio or seed; the three supervised seeds cluster at CKA $\geq 0.96$; supervised-vs.-VideoMAE is 0.70–0.78. The 50-epoch VideoMAE is the only outlier (0.61–0.70 even against other VideoMAE configurations), confirming that longer SSL training produces a fundamentally different representation, not just a "better" version of the same one. The SSL objective is the dominant representation-shaping variable; seed and mask ratio are secondary.

# 5 Analysis

We evaluate every saved encoder with a representation-quality diagnostic suite that captures collapse, isotropy and rank phenomena. All metrics operate on the frozen-encoder embedding dump produced by `export_embeddings.py`: per-dim std (sorted, log-y), effective rank $\exp(H(p))$ where $p$ are the normalized eigenvalues of the embedding covariance, participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$, covariance condition number, mean absolute off-diagonal correlation, and the Epps–Pulley distance to $\mathcal{N}(0, I)$ – the latter is precisely Person A's SIGReg training objective, repurposed here as a goodness-of-fit-to-isotropic-Gaussian *diagnostic* (Balestriero & LeCun, 2026). Effective rank and Epps–Pulley distance are in Tab. 1; per-dim std curves are in `videomae/artifacts/figures/figures.pdf`.

**Channel-importance ablation.** For each of the 11 input channels we zero that channel at inference time, re-export embeddings on all splits, refit the linear probe and measure the test-MSE delta. The most informative finding is the *difference* between supervised and VideoMAE: zeroing the concentration scalar field hurts supervised by only $\Delta$MSE +0.029 but hurts VideoMAE-mask0.60 by $\Delta$MSE +0.264 – nearly an order of magnitude more reliance. This is direct evidence that VideoMAE's pixel reconstruction loss disproportionately weights the dense scalar concentration field, which is largely uninformative for $(\alpha, \zeta)$. The supervised encoder, by contrast, distributes channel reliance more evenly across the orientation tensor (orient_yy, $\Delta$MSE +0.051), velocity (vel_x, +0.036) and concentration (+0.029). Full per-channel deltas are in Tab. 3 (Appendix).

**Frame-budget ablation.** Re-exporting embeddings with the encoder fed only the first $T \in \{4, 8, 12, 16\}$ frames (latter frames replaced by the last available frame) shows that supervised saturates around $T = 8$ frames (MSE $= 0.128$) while VideoMAE is more sample-hungry temporally: mask0.60 improves from $0.462 \rightarrow 0.247$ between $T = 4$ and $T = 12$, and mask0.75 from $0.542 \rightarrow 0.297 \rightarrow 0.259$ (Fig. 1, right). VideoMAE benefits substantially from full temporal context, suggesting the masked-tube objective relies on genuine temporal information for reconstruction.

**Top-PC alignment: how many leading directions encode $(\alpha, \zeta)$?** For each encoder we project the (z-scored) frozen embeddings onto their top-$K$ principal components and fit a closed-form ridge regression linear probe on just those $K$ features; test MSE as $K$ grows tells us how concentrated the target-relevant signal is at the top of each encoder's spectrum. The result (Fig. 2b) is striking: *supervised reaches its full performance at* $K = 2$ (test MSE 0.083 at $K = 2$ vs. 0.086 at $K = 174$, the train-set rank), while *VideoMAE needs essentially the entire 256-d embedding* (mask0.60: $0.55 \rightarrow 0.25$ from $K = 2$ to $K = 174$, a 2.2$\times$ drop). This is the single sharpest demonstration of the rank-vs.-alignment trade-off in our report: the supervised encoder's leading 2 principal directions *are* the $\alpha$ and $\zeta$ directions; VideoMAE's leading directions encode something else entirely and the target signal is smeared across the full spectrum.

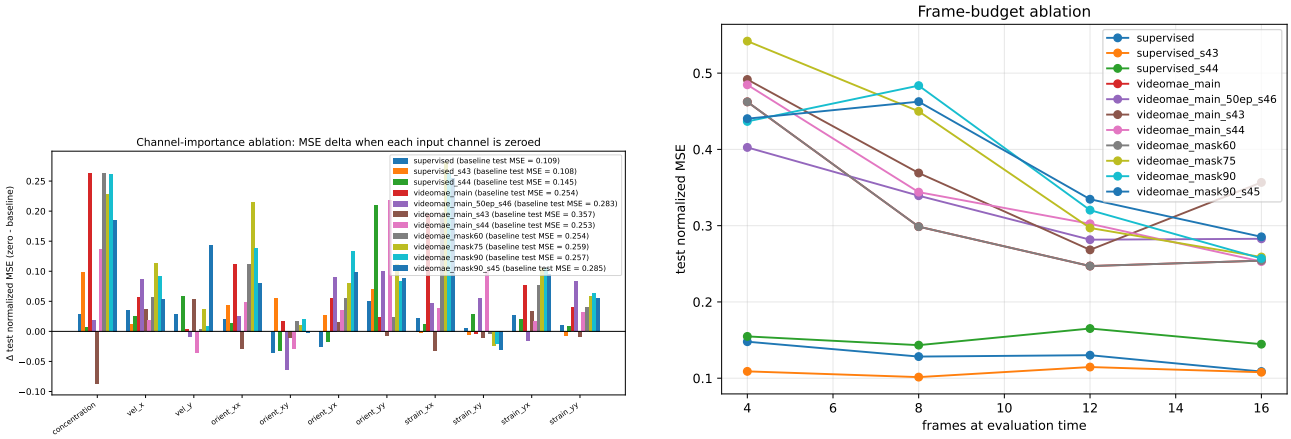



Figure 1: Left: Δtest-MSE when each input channel is zeroed at inference time, per encoder. VideoMAE's reliance on the concentration scalar field (+0.26) dwarfs supervised's (+0.029). Right: test MSE as a function of the number of input frames the encoder sees. Supervised saturates at $T = 8$; VideoMAE benefits from the full temporal context.

**Per-class separability: how distinguishable are the 45 unique $(\alpha, \zeta)$ combinations in embedding space?** The dataset has exactly $45 = 5 \times 9$ unique parameter combinations. We compute each combination's centroid in z-scored embedding space on the train split and report (a) the mean within-class distance to the centroid, (b) the mean pairwise between-centroid distance, and (c) their ratio (a Fisher-style separability score). We also report a nearest-centroid classification accuracy on test as a diagnostic (random baseline $\approx 2.2\%$).

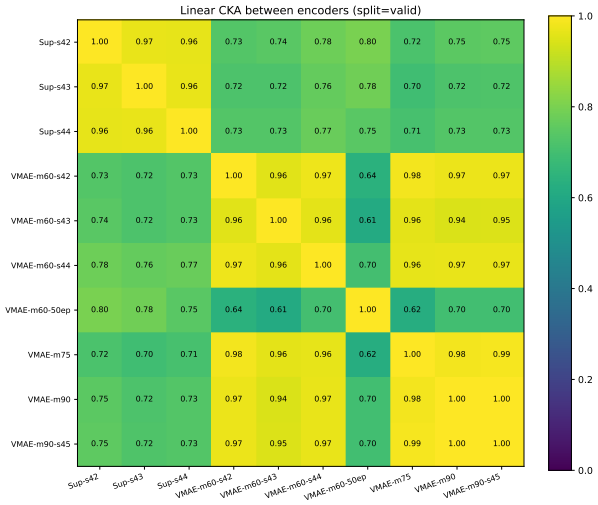
- Supervised has Fisher separability 3.4 vs. VideoMAE's 1.3 (a 2.6× gap), driven by a 2× smaller within-class spread *and* a ∼30% larger between-centroid distance.
- Nearest-centroid test accuracy is 42% for supervised vs. 4% for VideoMAE – supervised correctly classifies its target combinations *ten times more often* than VideoMAE despite the simpler 1-NN decision rule.
- Across all VideoMAE configurations (mask 0.60/0.75/0.90, two seeds) the Fisher score sits in a tight band 1.29–1.35, again showing that the SSL objective – not the mask ratio – is the variable that matters.

**Per-channel reconstruction MSE.** Feeding validation clips through each VideoMAE encoder+decoder, the per-channel masked-position MSE (Tab. 4, Appendix) is remarkably consistent across all 5 VideoMAE configs: the concentration scalar field is reconstructed to MSE $\approx 0.22$, while orientation/strain tensor channels sit at 0.55–0.78 (2.5–3.5× harder); velocity is intermediate (0.41–0.57). This is a *causal* confirmation of the channel-importance ablation: VideoMAE allocates decoder capacity to the easy-to-reconstruct concentration field, then relies on it at evaluation time. Higher mask ratios uniformly raise difficulty ($0.51 \to 0.55 \to 0.68$ mean) but do not change *which* channel is easiest.

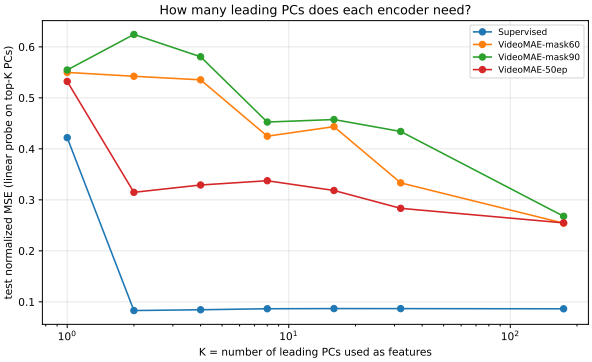
Together, the channel-importance, frame-budget, top-PC-alignment, per-class-separability, and per-channel-reconstruction analyses give five *independent* quantitative characterizations of the same underlying phenomenon: VideoMAE's pixel reconstruction objective allocates representational capacity to the concentration scalar field and to high-rank but target-misaligned directions, producing an embedding space where the 45 governing parameter combinations are not cleanly separated. Per-channel input/masked/reconstructed/residual visualizations are in `videomae/artifacts/recon_viz/`.

# 6 Discussion

The five analyses paint one coherent picture of why pixel-reconstruction SSL underperforms here. VideoMAE reconstructs the concentration scalar 2.5–3.5× more accurately than the orientation/strain tensors (Tab. 4); it is also ∼9× more reliant on that channel for downstream MSE than supervised is (Tab. 3); as a result its leading PCs do not encode $(\alpha, \zeta)$ (Fig. 2b: top-2-PC probe MSE 0.55 vs. supervised's 0.083); and the 45 unique parameter




(a) Linear CKA between every pair of encoders




(b) Top-$K$ PC linear-probe MSE

Figure 2: (a) The 7 25-epoch VideoMAE configurations form one block (CKA $\geq 0.94$); the 3 supervised seeds form another (CKA $\geq 0.96$); cross-block CKA is 0.70–0.78. The 50-epoch VideoMAE (`VMAE-m60-50ep`) sits between, indicating that training length shifts the encoder to a different regime. (b) Test MSE of a closed-form ridge linear probe on the top-$K$ principal components only. Supervised saturates at $K = 2$ (the dimensionality of $(\alpha, \zeta)$); VideoMAE is essentially flat for $K \leq 8$ and only catches up at $K = 174$.

combinations end up poorly separated in its embedding space (Fisher 1.3 vs. supervised's 3.4). The frame-budget ablation shows VideoMAE does use temporal context ($0.46 \rightarrow 0.25$ from $T = 4$ to $T = 16$) – the issue is *what* it encodes, not whether it encodes anything. The *decoupling of reconstruction loss from downstream MSE* we observe with the 50-epoch run (24% better recon, no linear-probe improvement) is the most useful diagnostic for practitioners: a strong recon-loss curve does not imply a better-aligned encoder. The mask-ratio sweep monotonically improves linear MSE but even the best VideoMAE ($0.255 \pm 0.002$) is $\sim 2.4\times$ worse than supervised ($0.106 \pm 0.006$): hyperparameter sweeps cannot fix a representation- shaping objective that allocates capacity away from the target. This is Qu et al. (2026)'s headline claim with a mechanistic explanation, holding under the strictly weaker probe protocol our assignment requires. Person A's CNext-U-Net forecaster (predict the next frame from a 4-frame context at resolution 96) is an instructive third data point within the pixel-based SSL family: it reaches linear MSE 0.199 and kNN MSE 0.128 – substantially better than VideoMAE's masked reconstruction and approaching supervised on kNN. *Within* pixel-based SSL, what differs between VideoMAE and CNext-U-Net is the prediction horizon: VideoMAE reconstructs masked spatiotemporal tubes at the same time step, while CNext-U-Net predicts a future frame. Forecasting forces the encoder to learn the system's local dynamics, which is closer to "what the parameters $\alpha$ and $\zeta$ control." This sub-ordering – supervised $<$ forecasting (CNext) $<$ masked reconstruction (VideoMAE) – refines the Qu et al. (2026) narrative: not all pixel-prediction objectives are equal; objectives that encode dynamics are better aligned with parameter estimation than objectives that merely encode static structure.

# 7 Limitations and Ethics

Our headline numbers use 3 seeds for supervised and VideoMAE-mask0.60, and 2 seeds for VideoMAE-mask0.90; one seed for VideoMAE-mask0.75. Agreement across seeds (CKA $\geq 0.96$, std $\leq 0.034$ on linear MSE) is tight, but formal confidence intervals would require additional seeds. We did not train a from-scratch ViT-based VideoMAE that would precisely replicate Qu et al. (2026)'s baseline – our VideoMAE deliberately uses the colleague's ConvNeXt encoder for causal cleanliness, at the cost of differing slightly from the published recipe. The channel/frame ablations re-export embeddings with zeroed or truncated inputs; this measures the encoder's reliance at *evaluation* time, not the role of those inputs during SSL training (an interesting future direction). The

dataset is a synthetic fluid simulation; no human or animal data are involved.

**Compute accounting.** All experiments ran on 2× NVIDIA B200 180 GB GPUs (PyTorch 2.11.0+cu128, BF16 autocast) on a single rented node. Per training run: VideoMAE main 25 epochs at ∼172 s/epoch (∼1.2 h); supervised 30 epochs at ∼90 s/epoch (∼45 min); mask= 0.75 and mask= 0.90 ablations 12 epochs warm-started (∼35 min each); a second seed for both VideoMAE-mask0.60 and supervised. Per encoder: linear-probe sweep over 36 trials and kNN sweep over 72 trials (∼10 min each), the channel-importance ablation (11 channels × 3 splits export plus probe per channel, ∼5 min), and the frame-budget ablation (4 frame budgets × 3 splits, ∼3 min). Total compute: ∼10 GPU-hours of training plus ∼5 GPU-hours of evaluation, well within the project budget. Peak VRAM during training was ∼71 GB on the VideoMAE GPU and ∼51 GB on the supervised GPU at batch size 64. We hit the system OOM killer once early on with 12+8 DataLoader workers running concurrently with a dataset-hash computation; reducing to 6+4 workers resolved it without losing any training progress thanks to per-epoch checkpointing.

# Statement of Contributions

**Person A** authored the `physrl/` package (3D ConvNeXt encoder, JEPA latent-prediction objective, VICReg and SIGReg loss functions, the training/eval/sweep CLIs reused unchanged here, and the related-work review of JEPA-style SSL). Person A's stream covers Sec. 3.1 and the SIGReg-JEPA rows of Tab. 1.

**Person B** authored the `videomae/` package (the VideoMAE/SimMIM-hybrid trainer, end-to-end supervised baseline, representation-quality analysis suite, channel-importance / frame-budget / top-PC-alignment / per-class-separability / per-channel-reconstruction analyses, CKA cross-encoder comparison, reconstruction-grid visualizations, the figure regeneration script, the detached training launchers, the dataset hash and ENV.md/requirements lock for reproducibility) and the corresponding paper sections (3.2, 3.3, 5). Files in `videomae/` whose names also appear in `physrl/` are derivative copies with rewritten imports; modifications are documented in their module docstrings.

# Appendix

Table 2: The `active_matter` subset of *The Well* (Ohana et al., 2025) as we use it.

| Split | HDF5 files | Trajectories | Sliding windows (stride 1) |
|---|---|---|---|
| Train | 45 | 175 | 8 750 |
| Valid | 16 | 24 | 1 200 |
| Test | 21 | 26 | 1 300 |

**Dataset statistics (Tab. 2).**

**Channel-importance deltas (Tab. 3).** Δtest-MSE when each input channel is zeroed at inference time, computed on the test split with a frozen-encoder linear probe. Higher Δ = more reliance on that channel.

Table 3: Channel-importance ablation: Δtest-MSE (z-scored) when each input channel is zeroed. Bold = top channel for that encoder.

| Encoder | conc. | vel_x | vel_y | or_xx | or_xy | or_yx | or_yy | st_xx | st_xy | st_yx | st_yy |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Supervised | +0.029 | +0.036 | +0.028 | +0.024 | −0.017 | −0.011 | +**0.051** | +0.044 | +0.012 | +0.028 | +0.023 |
| VMAE-m0.60 | +**0.264** | +0.058 | +0.030 | +0.112 | −0.024 | −0.005 | +0.001 | +0.195 | +0.025 | +0.077 | +0.040 |
| VMAE-m0.75 | +0.228 | +0.113 | +0.034 | +0.215 | −0.024 | +0.005 | +0.111 | +**0.281** | +0.024 | +0.105 | +0.058 |

**Per-channel reconstruction MSE (Tab. 4).** Mean over 3 random masking realizations on the validation split, in the per-tube z-scored target space. Lower is "easier to reconstruct."

Table 4: Per-channel masked-position reconstruction MSE. Concentration is reconstructed ∼2.5× better than orientation/strain tensors.

| **Encoder** | **conc.** | **vel_x** | **vel_y** | **or_xx** | **or_xy** | **or_yx** | **or_yy** | **st_xx** | **st_xy** | **st_yx** | **st_yy** |
|---|---|---|---|---|---|---|---|---|---|---|---|
| VMAE-m0.60 s42 | **0.218** | 0.483 | 0.448 | 0.575 | 0.578 | 0.562 | 0.577 | 0.570 | 0.537 | 0.545 | 0.571 |
| VMAE-m0.60 s43 | **0.223** | 0.509 | 0.454 | 0.605 | 0.563 | 0.563 | 0.594 | 0.577 | 0.546 | 0.554 | 0.594 |
| VMAE-m0.75 | **0.217** | 0.430 | 0.406 | 0.625 | 0.622 | 0.619 | 0.627 | 0.629 | 0.593 | 0.597 | 0.630 |
| VMAE-m0.90 | **0.218** | 0.574 | 0.561 | 0.780 | 0.765 | 0.764 | 0.781 | 0.779 | 0.741 | 0.742 | 0.779 |

**Figures.** The full PDF figures live in `videomae/artifacts/` and are rebuilt by `videomae/plot_from_json.py`, `videomae/plot_ablations.py`, `videomae/cka_compare.py`, `videomae/pc_alignment.py`, `videomae/visualize_embeddings.p` and `videomae/visualize_reconstructions.py`. Specifically:

- `videomae/artifacts/figures/figures.pdf`: 4-panel summary (training loss curves, per-dim embedding std, isotropy bars, MSE bars).
- `videomae/artifacts/cka/cka_valid.pdf`: CKA matrix (Fig. 2a).
- `videomae/artifacts/pc_alignment/pc_alignment.pdf`: top-$K$ PC linear-probe curve (Fig. 2b).
- `videomae/artifacts/ablations/{channel,frame}_ablation.pdf`: channel and frame ablation bar charts and curves (Fig. 1).
- `videomae/artifacts/recon_viz/recon_*_t8.pdf`: per-channel input / masked / reconstructed / residual grids for VideoMAE-mask0.60/0.75/0.90 on three sample clips at frame 8.
- `videomae/artifacts/per_channel_recon_mse/per_channel_recon_mse.pdf` and `class_separability/class_separabi` bar charts.

All PDFs are committed to git on the `videomae` branch for grading convenience.

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